Polynomial recurrence identities extend to all integer indices

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Abstract

We prove Conjecture 4.2 of Soykan's paper on negative-index Tribonacci matrix sequences. A polynomial expression in arithmetic subsequences of bilateral constant-coefficient recurrence sequences itself satisfies a recurrence with nonzero constant coefficient. Consequently, its vanishing on a terminal interval implies its vanishing everywhere. The proof makes explicit the extension of Bruckman's second-order argument suggested by Soykan; it uses standard finite-dimensional linear algebra.

Task 1: the conjecture and its scope

The source conjecture asks whether a polynomial identity in terms $H_{a_i n+b_i}$ of an arbitrary-order constant-coefficient recurrence, valid for all sufficiently large integers n, is valid for every integer n [1, Conjecture 4.2]. Here a_i, b_i are arbitrary integers. We use the intended bilateral convention: the recurrence is valid on $\mathbb{Z}$, with negative indices supplied by backward recurrence. The source's Theorem 1.1 explicitly uses a nonzero constant term for this extension. We prove the following stronger statement.

Theorem 1. *Let K be any field. For $1 \le i \le r$, let $H^{(i)} : \mathbb{Z} \to K$ satisfy a nonzero constant-coefficient homogeneous linear recurrence. Let $a_i, b_i \in \mathbb{Z}$ and $f \in K[X_1, \dots, X_r]$. Put*

$$u_n = f\big(H^{(1)}_{a_1 n+b_1}, \dots, H^{(r)}_{a_r n+b_r}\big).$$

If $u_n = 0$ for every sufficiently large integer n, then $u_n = 0$ for every integer n.

No distinct-root hypothesis is imposed. The slopes may be negative or zero. Taking all input sequences equal and taking integer coefficients for f proves the source conjecture. The harmless extra unused variable in its printed polynomial notation does not affect the assertion.

An invertible finite-dimensional representation

Let E denote the shift on bilateral sequences, $(EH)_n = H_{n+1}$. It is invertible. If $P(E)H = 0$ and $P(x) = x^t Q(x)$ with $Q(0) \neq 0$, then $Q(E)H = 0$ by applying E^{-t}. If Q is constant, H is zero; in that case use instead the annihilator $x - 1$. Otherwise rescale Q to be monic. Thus each input has an annihilator

$$Q_i(x) = x^{s_i} + c_{i,s_i-1}x^{s_i-1} + \cdots + c_{i,0}, \qquad s_i \ge 1, \quad c_{i,0} \neq 0.$$

For

$$v^{(i)}_m = (H^{(i)}_m, H^{(i)}_{m+1}, \dots, H^{(i)}_{m+s_i-1})^{\mathsf{T}},$$

the recurrence gives $v_{m+1}^{(i)} = M_i v_m^{(i)}$, where M_i is the companion matrix with superdiagonal ones and final row $(-c_{i,0}, \dots, -c_{i,s_i-1})$. Its determinant is $(-1)^{s_i} c_{i,0} \neq 0$. Consequently this state equation can be iterated in either direction.

Stack the selected states into

$$w_n = \big(v_{a_1 n+b_1}^{(1)}, \dots, v_{a_r n+b_r}^{(r)}\big)^{\mathsf{T}} \in K^m, \qquad m = \sum_{i=1}^{r} s_i.$$

Then, for every $n \in \mathbb{Z}$,

$$w_{n+1} = A w_n, \qquad A = \operatorname{diag}(M_1^{a_1}, \dots, M_r^{a_r}) \in \mathrm{GL}_m(K).$$

The shifts b_i affect w_0 only. There is a polynomial $F \in K[Z_1, \dots, Z_m]$ of degree at most $d = \max(0, \deg f)$ such that $u_n = F(w_n)$; for $f = 0$ choose $d = 0$.

Lemma 1. *For $A \in \mathrm{GL}_m(K)$ and F of degree at most d, the sequence $F(A^n w_0)$ satisfies a monic recurrence of order $D = \binom{m+d}{d}$ with nonzero constant coefficient, valid on $\mathbb{Z}$.*

Proof. On the vector space $V = K[Z_1, \dots, Z_m]_{\le d}$ define $TP = P \circ A$. This space has the monomials of total degree at most d as a basis, so has dimension D. The operator T is invertible: its inverse is $P \mapsto P \circ A^{-1}$. Write its characteristic polynomial as

$$R(x) = x^D + q_{D-1} x^{D-1} + \cdots + q_0.$$

Its constant coefficient is $q_0 = (-1)^D \det T \neq 0$. Cayley–Hamilton gives $R(T)F = 0$ as a formal polynomial. Evaluating at $w_n = A^n w_0$ gives

$$u_{n+D} + q_{D-1} u_{n+D-1} + \cdots + q_0 u_n = 0 \qquad (n \in \mathbb{Z}). \tag{1}$$

This argument concerns formal polynomials and also works over finite fields. □

Proof of Theorem 1. Choose an integer L for which $u_n = 0$ whenever $n \ge L$. Equation (1) at $n = L - 1$ gives $q_0 u_{L-1} = 0$, so $u_{L-1} = 0$. Apply the same step successively at $L-2, L-3, \dots$. Every integer is reached after finitely many steps. □

Corollary 1. *In the notation above, D consecutive zero values of u already imply $u_n = 0$ for all integers n. For a single input recurrence of order k and r selected subsequences, one may use $D = \binom{rk+d}{d}$ when its constant coefficient is nonzero.*

Proof. The leading coefficient in (1) is one, so the zero block propagates forward. Its nonzero constant coefficient propagates it backward. □

An extra input $(-1)^n$, as in the older second-order theorem, is allowed: it is itself a bilateral recurrence sequence. Repeated characteristic roots require no separate limiting argument.

Attribution, checks, and proposed follow-up tasks

Bruckman's solution of Rabinowitz's second-order problem already uses closure under polynomial expressions and backward propagation [2]. Soykan explicitly suggests extending that proof. The contribution here is a self-contained, arbitrary-order formulation and proof of the stated conjecture, with the standard argument and its antecedent credited; no new recurrence technique is claimed.

The accompanying exact verifier constructs polynomial pullback matrices, checks their inverses and Cayley–Hamilton identities, and compares the resulting recurrences with direct bilateral sequence evaluations. Cases include repeated roots, zero and negative slopes, and degree zero. These are implementation checks supporting the proof, not a proof by finite sampling. The source and bounded public-claim audit, including its access limits, are recorded separately. Review is by the authoring agent.

The following are four proposed tasks; they are not additional solved MathDB problems and have not been established here.

2. Determine the smallest uniform zero-block bound for fixed recurrence order, polynomial degree, and list of slopes, accounting for relations between the sampled states.

3. Design an exact algorithm that outputs the minimal annihilator of a supplied polynomial recurrence expression, together with a certificate, and bound its arithmetic complexity in terms of the sparse input size.

4. Decide whether the tail-to-all-integers implication continues to hold over characteristic zero when each affine index $a_i n + b_i$ is replaced by an integer-valued polynomial in n; give a proof or a counterexample.

5. For constant-coefficient recurrences over a commutative ring whose last coefficient need not be a unit, characterize when a bilateral sequence that is eventually zero must vanish identically.

References

[1] Y. Soykan, *Tribonacci and Tribonacci-Lucas Matrix Sequences with Negative Subscripts*, Communications in Mathematics and Applications **11**(1) (2020), 141–159, doi:10.26713/cma.v11i1.1103. Primary manuscript: arXiv:1809.09507v1.

[2] P. S. Bruckman, solution to problem H-487 proposed by S. Rabinowitz, The Fibonacci Quarterly **33**(4) (1995), 382.